

HOME POWER SAVERS DEMYSTIFIED

Are power savers really worth their salt or just a hoax? When and where are they better suited? Find answers to these and many other questions

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The article 'Do Power Savers Really Save Power?' published in March issue of EFY leaves many questions unanswered, such as:

1. Do most of these tiny gadgets contain PF capacitors of proper value, kVAR rating and appropriate type (low-loss, self-healing)?

2. Does a single PF capacitor offset the effect of all types of varying household loads—low and high, resistive (comprising lights, computer, TV, heater and geyser) and inductive (fans, fridge, air-conditioner and pump motor)?

3. Is its use really legal? Are the select customer testimonies produced

by the manufacturer any real proof of the effectiveness of the item? Is a 30-day warranty good enough? Are there any reputed manufacturers of home power saver devices? Why are these gadgets better suited for 3-phase loads?

4. If, in case, the inductive equipment used in premises have a high BEE (Bureau of Energy Efficiency) star rating, will the power saver have much effect?

Before we find answers to these questions, note that electricity authorities do not charge consumers on the basis of apparent power (kVAR) but true power (kW) consumption. Depending on the load, putting extra capacitors across mains may or may not

improve power factor. In certain cases like when your load is almost resistive, your power factor may degrade and your electricity bill may go up.

Improved power factor has some obvious advantages like reduction in heating losses in transformers, longer equipment life and enhanced capacity of electrical system/equipment.

PF capacitors specifications

PF capacitors used for direct insertion in mains line are generally quite large. Their values may range from a few tens to hundreds of microfarads with voltage rating exceeding 240V AC (for single-phase) to 440V AC (for phase-to-phase in 3-phase systems).

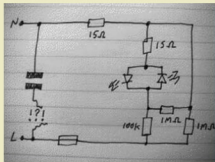
For exact calculations, you may refer my article 'Power Factor Correction' published in April 2006 issue of EFY. It provides a theoretical background on the usage of specific type, value and voltage rating of PF capacitors, to be switched into



(a) Physical appearance (plugged into mains)



(b) Inside contents (PF capacitor, resistors and LEDs)



(c) Circuit (undecoded part probably relates to switching arrangement for PF capacitor)

Fig. 1: Typical power saver